SPEECH & LANGUAGE PROCESSING · PROJECT REPORT

HyperBot

A voice-enabled chatbot for supercar & hypercar information, built with speech recognition and a deep-learning (BiLSTM) intent classifier

Live application	https://kanavbh29.github.io/slp/
Source code	https://github.com/KanavBh29/slp
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Course / Instructor	
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1. Abstract

HyperBot is a publicly hosted, voice-enabled chatbot that answers questions about 35 supercars and hypercars: their details, specifications, prices and where to buy them. The user's speech is transcribed in the browser by the Web Speech API, or by an on-device Whisper model when the browser's speech service is unavailable. Car names, brands and budgets are extracted with exact, fuzzy and phonetic matching, and replaced by placeholder tokens. A **Bidirectional LSTM** initialised with pre-trained **GloVe** embeddings then classifies the sentence into one of **26 intents**. A dialogue manager with context memory and slot filling builds the answer from a curated knowledge base, shows it next to the recognised speech, and reads it aloud.

The model was trained on a hand-built corpus of 796 utterance templates, augmented to 4,158 sentences. In 5-fold cross-validation on *phrasings never seen during training*, it reaches **81.5% ± 3.3** accuracy. That is 4.5 points above the same network trained without transfer learning, and on par with a bag-of-words MLP baseline that is twice its size. On the in-distribution random split it reaches 95.4%. The whole system runs client-side and is deployed to GitHub Pages by a CI pipeline that runs end-to-end tests before every release.

35	26	81.5%	<50 ms
cars from 18 brands	intents · 796 templates	accuracy on unseen phrasings (5-fold CV)	in-browser inference per message

2. Introduction & objectives

Car enthusiasts often want quick, specific answers ("How fast is the Koenigsegg Jesko?", "Where can I buy a McLaren 750S?") without searching through brochures and listings. A voice interface makes this natural, but it brings two NLP problems. First, speech recognisers often mishear exotic model names. Second, users phrase the same question in many different ways. The project objectives were to:

1. accept voice input and convert it to text with a suitable speech recognition technique;
2. classify the text with a deep-learning intent model and generate an appropriate response;
3. display both the recognised speech and the chatbot response;
4. deploy the application online with a public link; and
5. document the dataset, architecture, methodology and results.

3. System architecture

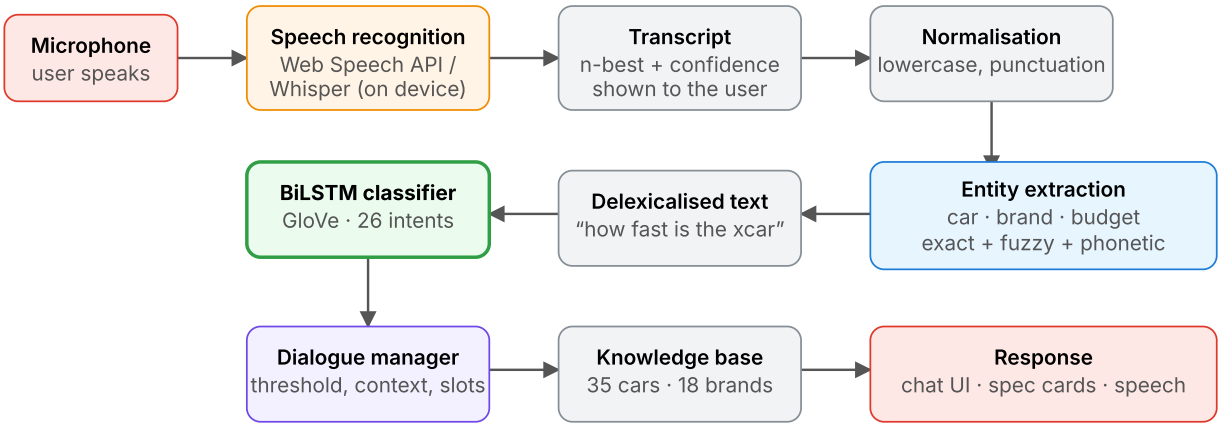


Figure 1. End-to-end pipeline. Every stage runs inside the user's browser except the optional cloud ASR of the Web Speech API.

The application is a static web app: HTML, CSS and JavaScript ES modules, a JSON knowledge base, and the exported model weights. The neural network is trained offline in Python with TensorFlow/Keras, and its weights are exported to a compact binary file. A hand-written JavaScript implementation of the forward pass runs inference in the browser. This design needs no server, has no API keys and no cold starts, and can be hosted free on GitHub Pages.

4. Speech recognition

Engine	Technique	Runs	Notes
Web Speech API (default)	Browser vendor's neural ASR (Google's speech service in Chrome)	Browser + vendor cloud	Streaming interim results; n-best list (3 alternatives) with confidence; en-US/GB/IN/AU
Whisper-tiny.en (fallback)	Transformer encoder-decoder (39M params) via Transformers.js / ONNX Runtime Web	Fully on device	Used automatically if the cloud service is unreachable (e.g. Brave); push-to-talk; audio resampled to 16 kHz mono

General-purpose ASR is weakest on exactly the words this domain depends on: "Chiron" is heard as *"sheeran"*, "Huayra" as *"wire a"*, "Nevera" as *"never a"*. Three mechanisms compensate:

- **n-best rescoring:** a lower-ranked alternative is chosen if it is the only one that mentions a known car.
- **Fuzzy matching:** windows of 1–3 words are matched to every alias by normalised Levenshtein similarity (≥ 0.8). Only windows containing an uncommon word are considered, to avoid false positives.
- **Phonetic matching:** a simplified Metaphone key maps `sheeran` and `chiron` to the same key `xrn`, and `hurricane` and `huracan` to `hrkn`.

The recognised text and its ASR confidence are shown above each answer, so the user can always see what the system heard.

5. Dataset

5.1 Intent corpus

No public dataset exists for a supercar assistant, so a domain corpus was written by hand: **796 templates over 26 intents**. It includes short follow-ups ("how much does it cost"), unpunctuated speech-style text and an `out_of_scope` class. Templates contain slot placeholders `{car}`, `{brand}` and `{num}`, which are **delexicalised** to the tokens `xcar`, `xbrand` and `xnum`. As a result, the network learns the *intent* rather than memorising car names, and new cars can be added without retraining.

Augmentation (12 variants per template):

- polite prefixes and suffixes ("hey ...", "... please");
- unknown-entity simulation: 12% of car slots are filled with out-of-vocabulary tokens;
- ASR word-dropout: one non-entity word is deleted with probability 0.15.

The final training set has 4,158 sentences.

Intent	Meaning	Templates	Example	F1*
`greeting`	Opening the conversation	38	*"hey"*	0.77
`goodbye`	Ending the conversation	36	*"see you"*	0.75
`thanks`	Gratitude / acknowledgement	33	*"thanks a lot"*	0.60
`help`	What can the bot do	31	*"how can you help me"*	0.75
`bot_identity`	Questions about the bot	26	*"what is your name"*	0.44
`car_overview`	General details about a car	46	*"info on the Chiron"*	0.68
`car_specs`	Full specification sheet	29	*"Chiron specifications"*	0.96
`car_engine`	Engine / powertrain	33	*"what is the engine in the Chiron"*	0.93
`car_power`	Horsepower and torque	30	*"what is the horsepower of the Chiron"*	1.00
`car_top_speed`	Maximum speed	29	*"Chiron top speed"*	0.87
`car_acceleration`	0-100 km/h / 0-60 mph	29	*"Chiron 0 to 60"*	0.73
`car_price`	Price / cost / value	37	*"Chiron price"*	0.91
`where_to_buy`	Dealers, marketplaces, auctions	41	*"how can i buy a Chiron"*	0.78
`car_production`	Units built, availability	25	*"is the Chiron limited edition"*	0.62
`compare`	Head-to-head comparison	32	*"Chiron versus Jesko"*	0.92
`fastest_car`	Ranking by top speed	25	*"fastest hypercar"*	1.00
`quickest_car`	Ranking by acceleration	22	*"fastest 0 to 60 car"*	0.75
`most_powerful`	Ranking by power	21	*"most powerful hypercar"*	1.00
`most_expensive`	Ranking by price (desc)	19	*"which car costs the most"*	0.93
`cheapest`	Ranking by price (asc)	21	*"most affordable supercar"*	1.00
`budget_recommend`	Recommendations for a budget	34	*"best car under $400k"*	0.92
`electric_hybrid`	EV / hybrid listing	27	*"show me electric supercars"*	0.87

`list_cars`	Database contents	27	*"which cars can you tell me about"*	0.92
`brand_info`	Manufacturer information	27	*"Ferrari models"*	0.92
`definition`	Supercar vs hypercar	24	*"difference between a supercar and a hypercar"*	0.91
`out_of_scope`	Off-topic (politely declined)	54	*"what time is it"*	0.53

* F1 on fold 1 of the template cross-validation, i.e. on phrasings unseen during training.

5.2 Knowledge base

`web/data/cars.json` holds 35 cars (21 hypercars, 14 supercars; 2 electric, 17 hybrid) from 18 manufacturers. Each entry records the engine, power, torque, 0–100 km/h time, top speed, approximate price, production volume, availability status, a description, and aliases (including common mishearings). The *where to buy* answer depends on the car's status:

- **available:** the official dealer and configurator;
- **limited:** the manufacturer, plus the secondary market;
- **sold out** or **discontinued:** specialist marketplaces (duPont REGISTRY, JamesEdition) and auction houses (Bring a Trailer, RM Sotheby's).

Car	Class	Powertrain	hp	km/h	0-100 s	Price	Status
Aston Martin Valhalla	Hypercar	Hybrid	1,064	350	2.5	$1.0M	limited
Aston Martin Valkyrie	Hypercar	Hybrid	1,139	350	2.6	$3.2M	sold out
Bugatti Tourbillon	Hypercar	Hybrid	1,775	445	2.0	$4.1M	sold out
Bugatti Chiron Super Sport	Hypercar	Petrol	1,578	440	2.4	$3.9M	sold out
Bugatti Veyron 16.4	Hypercar	Petrol	987	407	2.5	$1.8M	discontinued
Chevrolet Corvette ZR1	Supercar	Petrol	1,064	375	2.4	$175k	available
Czinger 21C	Hypercar	Hybrid	1,250	407	1.9	$2.0M	limited
Ferrari F80	Hypercar	Hybrid	1,184	350	2.15	$3.9M	sold out
Ferrari 12Cilindri	Supercar	Petrol	819	340	2.9	$423k	available
Ferrari 296 GTB	Supercar	Hybrid	819	330	2.9	$340k	available
Ferrari SF90 Stradale	Supercar	Hybrid	986	340	2.5	$524k	available
Ferrari LaFerrari	Hypercar	Hybrid	950	350	2.4	$4.0M	discontinued
Ford GT	Supercar	Petrol	660	348	3.0	$1.0M	discontinued
Gordon Murray T.50	Hypercar	Petrol	654	362	2.8	$3.1M	sold out
Hennessey Venom F5	Hypercar	Petrol	1,817	500	2.6	$2.1M	sold out
Koenigsegg Gemera	Hypercar	Hybrid	2,270	400	1.9	$1.7M	sold out
Koenigsegg Jesko Absolut	Hypercar	Petrol	1,600	500	2.5	$3.0M	sold out
Lamborghini Temerario	Supercar	Hybrid	907	343	2.7	$340k	available
Lamborghini Revuelto	Supercar	Hybrid	1,001	350	2.5	$608k	available
Lamborghini Huracán STO	Supercar	Petrol	631	310	3.0	$330k	discontinued
Lamborghini Aventador SVJ	Supercar	Petrol	759	350	2.8	$550k	discontinued

Lotus Evija	Hypercar	Electric	1,973	350	2.9	$2.1M	limited
Maserati MC20	Supercar	Petrol	621	325	2.9	$250k	available
McLaren W1	Hypercar	Hybrid	1,258	350	2.7	$2.1M	sold out
McLaren 750S	Supercar	Petrol	740	332	2.8	$324k	available
McLaren Artura	Supercar	Hybrid	690	330	3.0	$250k	available
McLaren P1	Hypercar	Hybrid	903	350	2.8	$1.6M	discontinued
Mercedes-AMG ONE	Hypercar	Hybrid	1,049	352	2.9	$2.7M	sold out
Pagani Utopia	Hypercar	Petrol	852	350	3.0	$2.5M	sold out
Pagani Huayra	Hypercar	Petrol	720	383	3.2	$2.8M	discontinued
Porsche 911 Turbo S	Supercar	Hybrid	701	322	2.5	$272k	available
Porsche 911 GT3 RS	Supercar	Petrol	518	296	3.2	$241k	available
Porsche 918 Spyder	Hypercar	Hybrid	887	345	2.6	$1.7M	discontinued
Rimac Nevera	Hypercar	Electric	1,914	412	1.85	$2.2M	limited
SSC Tuatara	Hypercar	Petrol	1,750	455	2.5	$1.9M	limited

Specs are manufacturer claims. Prices are approximate US base prices for new cars, or typical market values for sold-out and discontinued cars.

6. Text preprocessing & entity extraction

1. **Normalisation:** lowercase, remove apostrophes ("what's" → "whats"), replace punctuation with spaces while keeping decimals and model names such as "t.50".
2. **Entity extraction:** car aliases are matched longest first (so "Bugatti Chiron" wins over "Bugatti"), then fuzzy and phonetic car matches, then brands. Matches are replaced by `xcar` or `xbrand`.
3. **Number masking:** amounts such as "$2.5 million", "300k" and "2.5s" become `xnum`, while names like "V12" are kept. A separate parser converts budgets to USD, including spoken forms ("two million", "half a million", "five hundred thousand") and rupees ("3 crore").
4. **Encoding:** tokens are mapped to vocabulary ids (1 = OOV) and post-padded to length 24.

The Python (training) and JavaScript (inference) preprocessors are tested against each other on tricky inputs to guarantee identical tokens.

7. Model architecture & training

Layer	Output	Parameters	Notes
Input	24 token ids	0	0 = padding (masked)
Embedding	24 × 50	405,800	GloVe-50d init (Wikipedia + Gigaword, 400k words), fine-tuned, `mask_zero`
Bidirectional LSTM	128	58,880	64 units per direction
Dropout	128	0	rate 0.3
Dense	64	8,256	ReLU

Dropout	64	0	rate 0.3
Dense (output)	26	1,690	softmax
Total		**474,626**	

Why a BiLSTM? Intent depends on word order and context, e.g. "is the *xcar* faster than the *xcar*" (compare) versus "fastest car" (ranking). A bidirectional LSTM reads the sentence in both directions and handles variable length through masking.

Why GloVe transfer learning? The vocabulary (8,116 words) contains every training word plus the 8,000 most frequent English words, all initialised with GloVe vectors. Words that never appear in training receive no gradient and keep their pre-trained meaning. This lets the model handle unseen wording, e.g. "where can I **acquire** one" → where_to_buy and "which **vehicles** are battery electric" → electric_hybrid. The placeholder tokens start from the vectors of "supercar", "manufacturer" and "million".

Training: Adam (learning rate 2·10-3), sparse categorical cross-entropy, batch size 32, up to 80 epochs, early stopping on validation loss (patience 12, best weights restored), 10% validation split, seed 42. The final model trained for 38 epochs and reached **98.8% validation accuracy**.

Browser export: embeddings are quantised to int8 with a per-row scale, and the other weights stay float32, giving a ≈0.7 MB binary plus an 80 KB vocabulary. The JavaScript LSTM (Keras gate order i, f, c, o) reproduces the Keras probabilities to within **5·10-7**.

8. Dialogue management

- **Confidence threshold (0.5):** below it, rule-based fallbacks use the extracted entities (two cars → compare, one car → overview, a budget → recommendations). Otherwise the bot asks the user to rephrase.
- **Context memory:** the most recent cars are remembered, so "How fast is it?", "Price please" and "Compare it with the Jesko" work. Context is reused only for genuine follow-ups (a pronoun or a very short question). "What is the price of the Tesla Roadster?" gets "I don't have that car" instead of a wrong answer.
- **Slot filling:** "Recommend me a car" → "What is your budget?" → "500 thousand" → recommendations.
- **Response types:** text answers, spec cards, comparison tables (winners highlighted), top-5 rankings (optionally per brand), budget lists, purchase links, and text-to-speech output.

9. Implementation & deployment

Component	Technology
Model training & evaluation	Python 3, TensorFlow 2 / Keras, NumPy, Matplotlib, GloVe embeddings
Frontend	HTML5, CSS, vanilla JavaScript ES modules (no framework), responsive for mobile
Speech	Web Speech API (SpeechRecognition + SpeechSynthesis); Transformers.js Whisper fallback
Inference	Custom JavaScript BiLSTM forward pass over exported weights

Testing	Node.js test scripts (model parity, tokenizer parity, entities, budgets, 32-turn dialogue); Playwright browser tests
Hosting / CI-CD	GitHub Actions → GitHub Pages (tests must pass before deploy)

```
web/                    deployed static site
  js/speech.js          speech recognition engines + text-to-speech
  js/nlp.js             normalisation, entity extraction, budget parser
  js/model.js           BiLSTM forward pass in JavaScript
  js/chatbot.js         dialogue manager + response generation
  data/cars.json        knowledge base
  model/model.json|bin  vocabulary + weights
training/               intents.py, nlp.py, train.py, artifacts/
tests/                  parity, entity and conversation tests
.github/workflows/      test & deploy pipeline
```

10. Results & evaluation

10.1 Intent classification

Three neural models were compared under two protocols:

- **Random split** (85/15 of the generated sentences, 3,547/626). This is in-distribution: test sentences can share a template with training sentences.
- **Template 5-fold cross-validation.** Each fold holds out 20% of *every intent's templates*, so every test sentence is a phrasing the model has never seen. This is the more realistic estimate for new users.

Model	Trainable params	Random: acc	Random: F1	Template CV: acc	Template CV: F1
Bag-of-words MLP (baseline)	1.04 M	97.1%	0.930	80.6% ± 1.1	0.789 ± 0.028
BiLSTM, embeddings from scratch	0.47 M	94.9%	0.910	77.0% ± 1.3	0.740 ± 0.009
BiLSTM + GloVe (deployed)	0.47 M	95.4%	0.918	81.5% ± 3.3	0.792 ± 0.044

Accuracy per fold	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean
BoW-MLP	79.6%	82.1%	80.5%	79.4%	81.4%	**80.6%**
BiLSTM (scratch)	75.3%	77.5%	77.3%	79.0%	75.8%	**77.0%**
BiLSTM + GloVe	83.9%	84.6%	83.8%	79.0%	76.1%	**81.5%**

Findings.

- **Transfer learning helps most on unseen phrasings.** GloVe initialisation adds +4.5 accuracy points and +0.052 macro-F1 over the same BiLSTM trained from scratch.
- **On the in-distribution split, all models exceed 94%.** The bag-of-words MLP is slightly ahead there because keyword memorisation works well when templates repeat.
- **On genuinely new phrasings, the GloVe-BiLSTM has the highest mean accuracy and F1.** The gap to the baseline is within fold-to-fold variance, so the two are best described as comparable. The BiLSTM is less than half the size and also models word order.

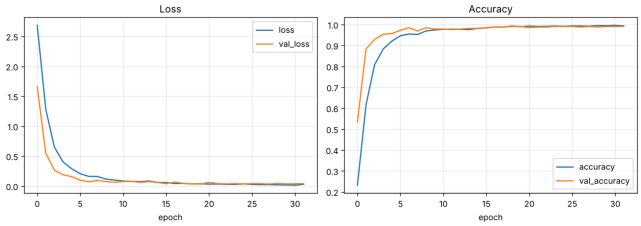


Figure 2. Loss and accuracy curves (fold 1).

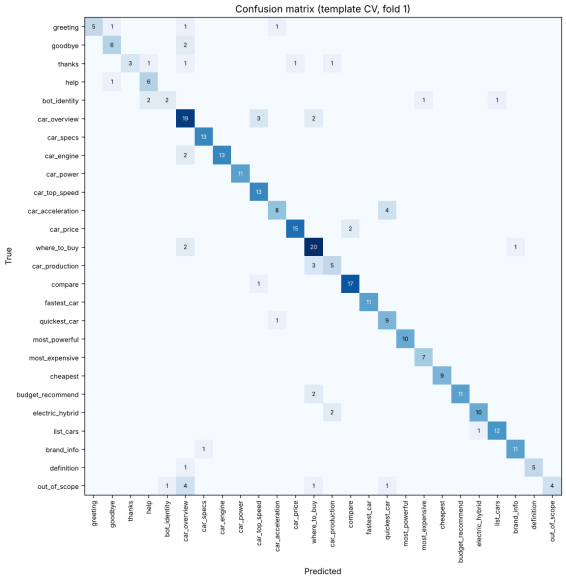


Figure 3. Confusion matrix on held-out templates (fold 1).

Error analysis. Most remaining errors fall into four groups:

- one-word slang with no embedding ("thx", "cya", "yo");
- off-topic questions shaped like "who/what is X" ("who are the beatles" → car_overview);
- a single car's acceleration confused with the acceleration ranking ("zero to sixty" → quickest_car);
- overlap between availability and purchasing ("is it sold out" → where_to_buy).

The dialogue manager limits the impact of these errors. For example, a car-specific intent with no car mentioned produces a clarifying question rather than a wrong fact.

10.2 End-to-end testing

Test (run in CI before every deployment)	Result
JavaScript BiLSTM vs Keras on 11 reference sentences	identical labels; max |Δp| = 5·10-7
Python vs JavaScript tokenizer on 13 tricky inputs	identical
Entity extraction incl. ASR errors ("bugatti sheeran", "koenig seg jesko", "rimac never a")	11 / 11 correct
Budget parser ("$300k", "2.5 million dollars", "two million", "half a million", "3 crore rupees")	8 / 8 correct

32-turn scripted conversation (follow-ups, comparisons, slot filling, out-of-scope)	32 / 32 intents and entities correct
Headless Chromium, desktop + mobile, simulated microphone	speech shown with confidence, correct answer, no console errors, no horizontal overflow
Simulated speech-service network failure	automatic switch to on-device Whisper with explanation

10.3 Sample dialogue (actual system output)

User (recognised speech)	Intent	HyperBot response (spoken version)
“Hello”	`greeting` 99.9%	Hello! Which supercar or hypercar are you curious about today?
“Tell me about the Bugatti Chiron”	`car_overview` 100.0%	The Bugatti Chiron Super Sport is a 2021 hypercar from Bugatti. The long-tail, top-speed focused Chiron. Its sibling was the first production car past 300 mph (490 km/h).
“How fast is it?”	`car_top_speed` 100.0%	The Bugatti Chiron Super Sport has a top speed of 440 km/h (273 mph).
“How much does it cost?”	`car_price` 100.0%	The Bugatti Chiron Super Sport costs about 3.9 million dollars.
“Where can I buy one?”	`where_to_buy` 100.0%	The Bugatti Chiron Super Sport is sold out, so you'd have to buy one second-hand, through a specialist like duPont Registry, JamesEdition, or an auction house like RM Sotheby's. Bugatti dealers also broker resales.
“Compare it with the Koenigsegg Jesko”	`compare` 100.0%	The Bugatti Chiron Super Sport wins on torque and 0–100 km/h, while the Koenigsegg Jesko Absolut wins on power, top speed and price.
“Recommend me a car”	`budget_recommend` 100.0%	Sure! What is your budget? For example “300 thousand dollars” or “2 million”.
“500 thousand”	`budget_recommend` 84.2%	With 500,000 dollars, my top pick is the Ferrari 12Cilindri at about 423,000 dollars.
“Is the Lamborghini Revuelto a hybrid?”	`car_engine` 100.0%	The Lamborghini Revuelto is powered by a 6.5L naturally aspirated V12 plug-in hybrid (hybrid powertrain), producing 1,001 hp.
“bugatti sheeran price”	`car_price` 100.0%	The Bugatti Chiron Super Sport costs about 3.9 million dollars.
“What is the price of the Tesla Roadster?”	`unknown_car` 100.0%	Sorry, I don’t have that car in my database yet. Ask me what cars I know to hear the full list.
“Tell me a joke”	`out_of_scope` 100.0%	I only know about supercars and hypercars. Try asking “What is the fastest car?”
“Thanks, bye”	`goodbye` 100.0%	Goodbye! Keep it pinned. 🏁

11. Application walkthrough

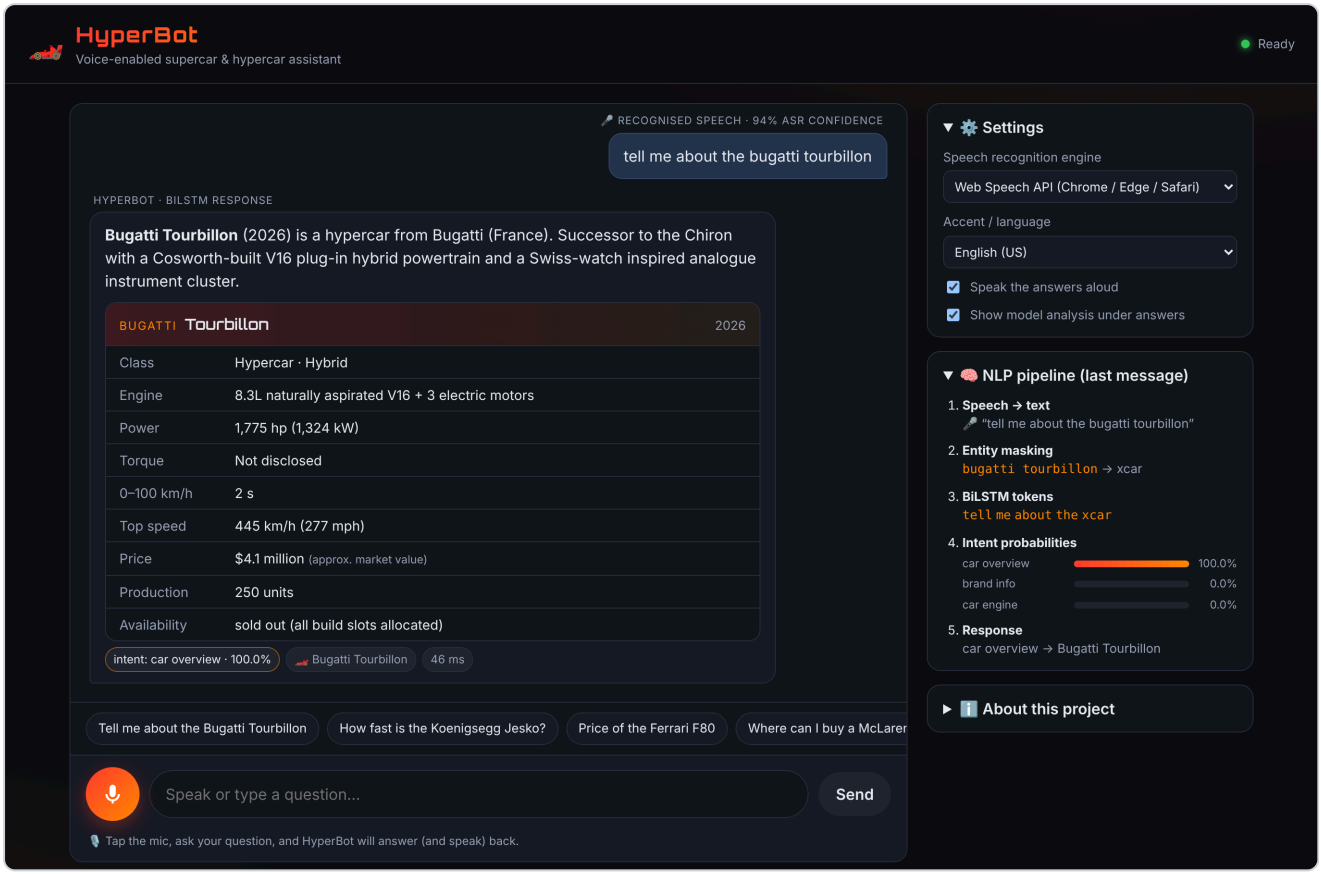


Figure 4. Voice query "tell me about the Bugatti Tourbillon": the recognised speech with 94% ASR confidence, the answer with a spec card, the predicted intent and confidence, and the NLP pipeline panel.

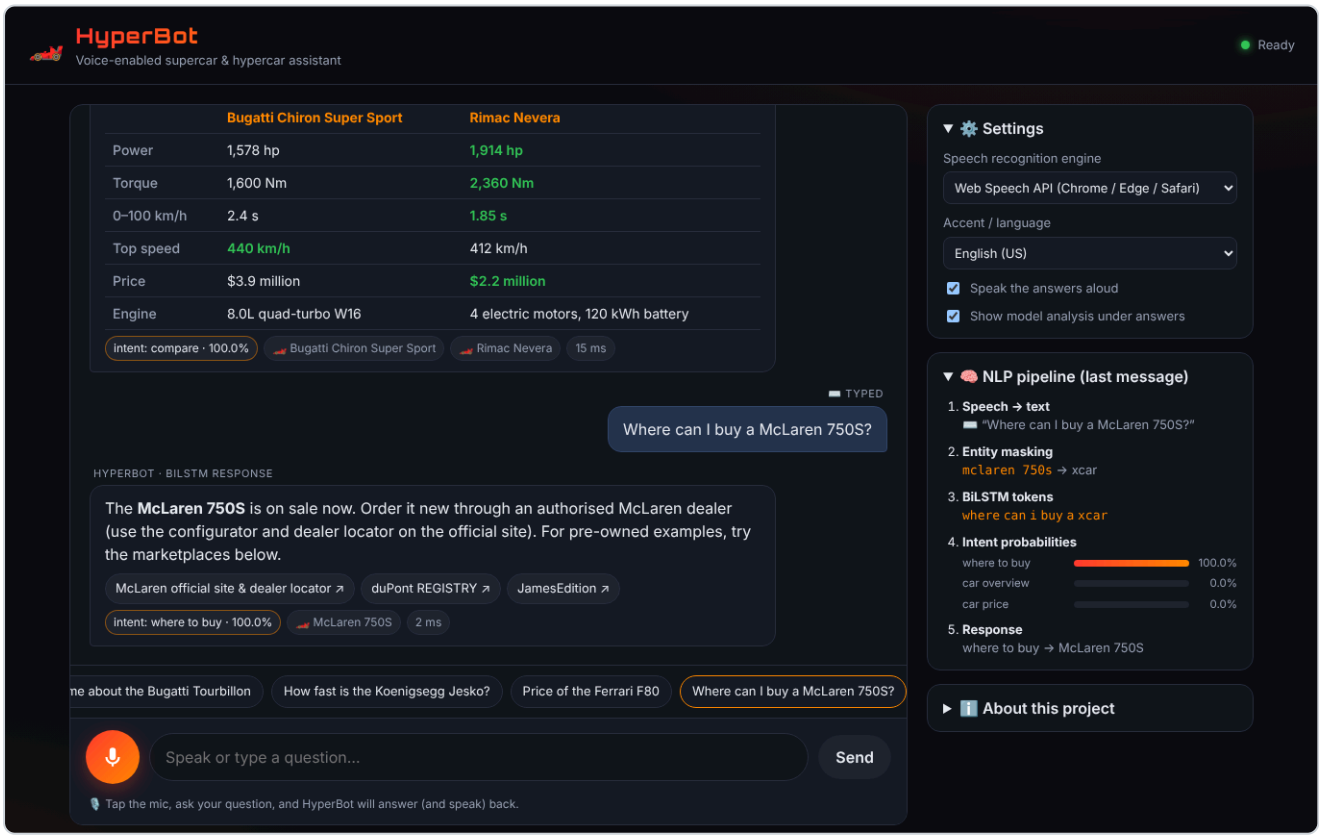


Figure 5. A comparison table with winners highlighted, and where-to-buy links for a car in production.

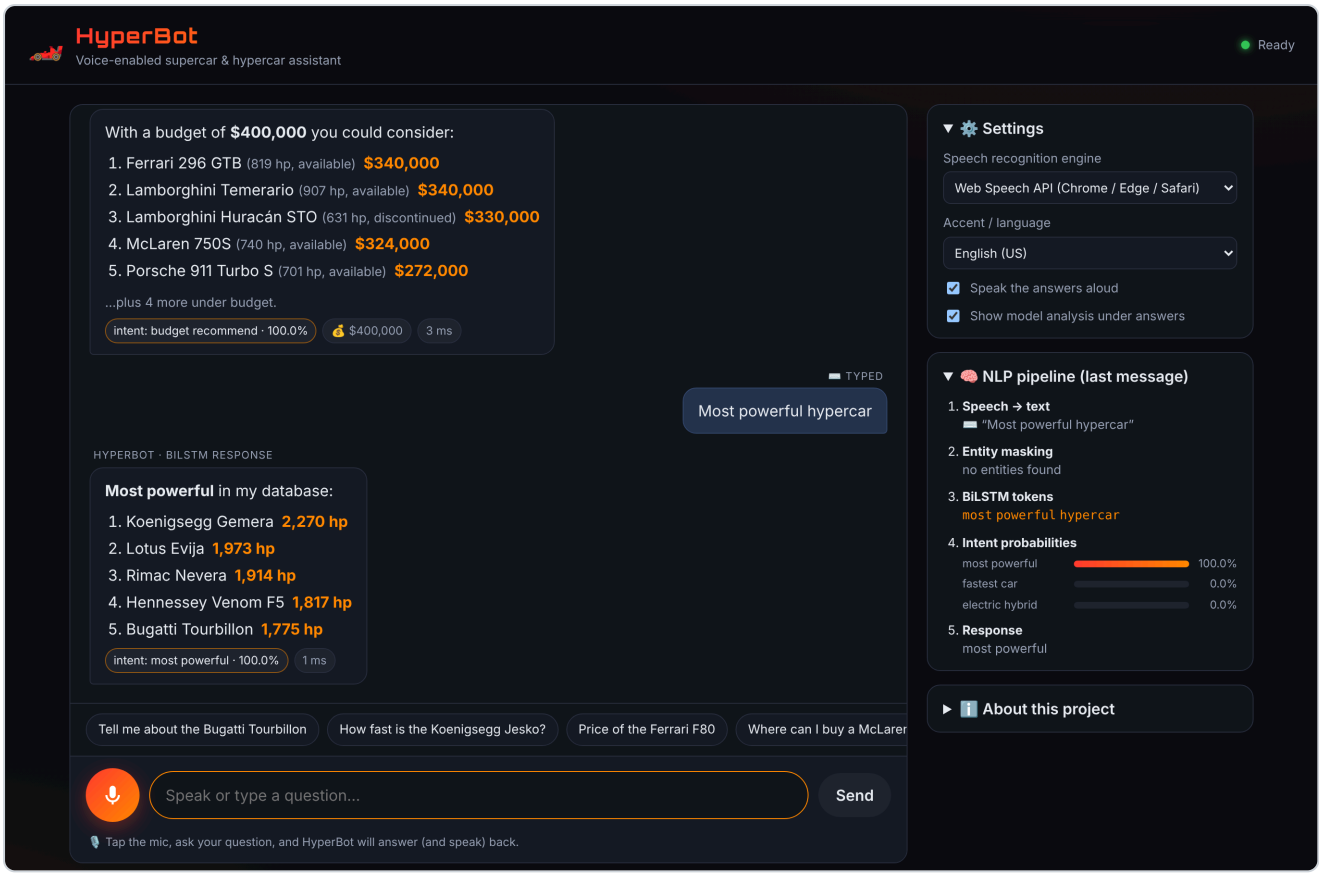


Figure 6. Budget recommendation (parsed $400,000) and a power ranking.

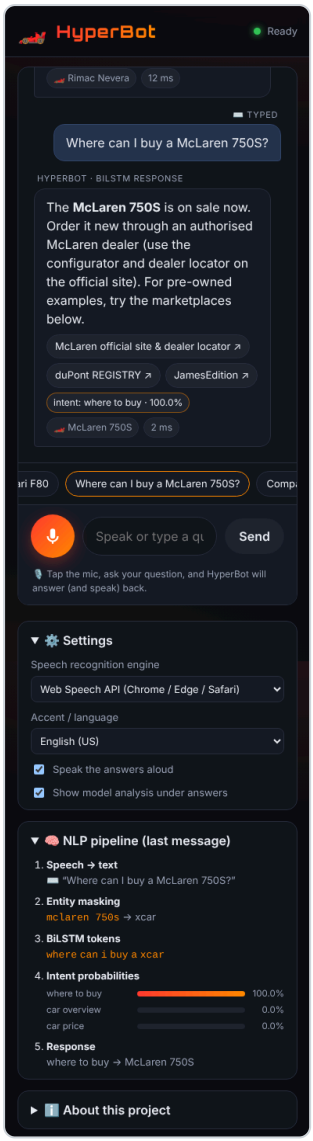


Figure 7. Responsive mobile layout.

Using the application

1. Open https://kanavbh29.github.io/slp/ in Chrome, Edge or Safari (HTTPS is required for the microphone).
2. Tap the red microphone button and allow microphone access.
3. Ask a question, for example "How much is the Ferrari F80?".
4. The recognised speech appears in the chat, followed by the answer, which is also spoken aloud.
5. Typing and the suggestion chips work as alternatives to voice.
6. In browsers that block the cloud speech service (e.g. Brave), the app switches to on-device Whisper. Tap the mic once to start and again to stop.

12. Limitations & future work

- The intent corpus is synthetic. Collecting real anonymised queries would give the largest robustness gain.
- Web Speech API quality and privacy depend on the browser vendor. Whisper keeps audio on the device but needs a ~40 MB download.
- Prices and availability change. The static JSON should be refreshed regularly or connected to a live listings API.
- Possible extensions: a fine-tuned transformer encoder (e.g. DistilBERT) for intents; joint intent + slot tagging (BiLSTM-CRF) instead of dictionary matching; multilingual ASR; images of each car.

13. Conclusion

HyperBot meets all project requirements. It accepts voice input, transcribes it with a speech recognition engine (with an on-device fallback), classifies intent with a deep-learning BiLSTM using transfer-learned GloVe embeddings, and answers questions about details, specifications, prices and

where to buy 35 supercars and hypercars. Each turn shows both the recognised speech and the response. The system is publicly deployed through an automated test-and-deploy pipeline. Evaluation on unseen phrasings shows a clear benefit from transfer learning (81.5% vs 77.0%). Domain-specific handling of speech recognition errors (n-best rescoring, fuzzy and phonetic matching) makes the voice experience reliable even for hard-to-pronounce car names.

14. References

1. S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, 9(8), 1997.
2. M. Schuster and K. K. Paliwal, "Bidirectional Recurrent Neural Networks," *IEEE Trans. Signal Processing*, 45(11), 1997.
3. J. Pennington, R. Socher and C. D. Manning, "GloVe: Global Vectors for Word Representation," *EMNLP*, 2014.
4. A. Radford et al., "Robust Speech Recognition via Large-Scale Weak Supervision (Whisper)," 2022.
5. W3C Community Group, "Web Speech API" specification.
6. V. I. Levenshtein, "Binary codes capable of correcting deletions, insertions, and reversals," *Soviet Physics Doklady*, 1966.
7. L. Philips, "Hanging on the Metaphone," *Computer Language*, 1990.
8. F. Chollet et al., Keras; M. Abadi et al., TensorFlow, 2015–2026.
9. Hugging Face, Transformers.js documentation.
10. Manufacturer press releases and official websites for vehicle specifications (see the knowledge base).

Appendix: reproducing the project

```
git clone https://github.com/KanavBh29/slp && cd slp
pip install -r training/requirements.txt
python3 training/train.py      # trains, evaluates (random split + 5-fold template CV),
                               # writes web/model/* and training/artifacts/*
npm test                       # parity, entity, budget and conversation tests
python3 -m http.server 8000 --directory web   # open http://localhost:8000
```